IGCSE 0607 Practice Questions — E7.1 — Combined Questions & Markscheme

Pythagoras' theorem

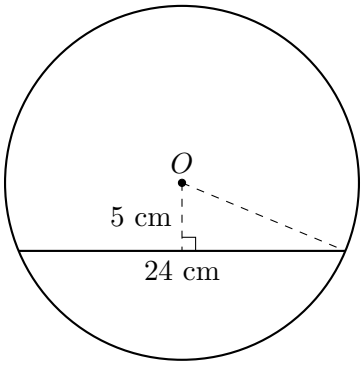
Total Questions: 18 — **Total Marks:** 113 Marks

Table of Contents & Quick Navigation

- **E7.1** Pythagoras' theorem
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

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1 A chord of a circle, centre O, has length 24 cm. The perpendicular distance from the centre of the circle to the chord is 5 cm.



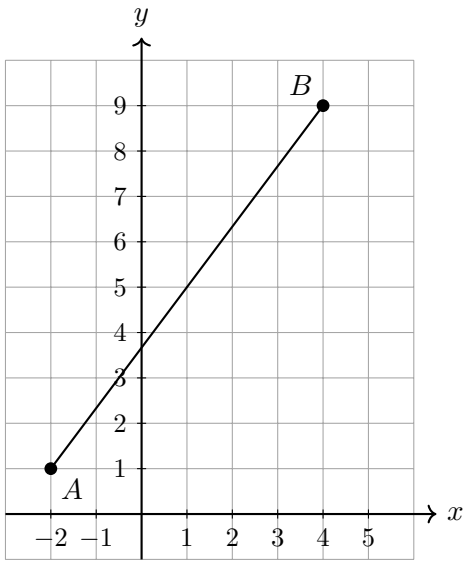
Calculate the radius of the circle.

..... cm **[3]**

2 The points $A(-4, 6)$ and $B(8, 1)$ lie on a coordinate grid.
Work out the exact length of the line segment AB.

..... **[3]**

3 Points $A(-2, 1)$ and $B(4, 9)$ are plotted on a coordinate grid.



Calculate the exact length of the line segment AB.

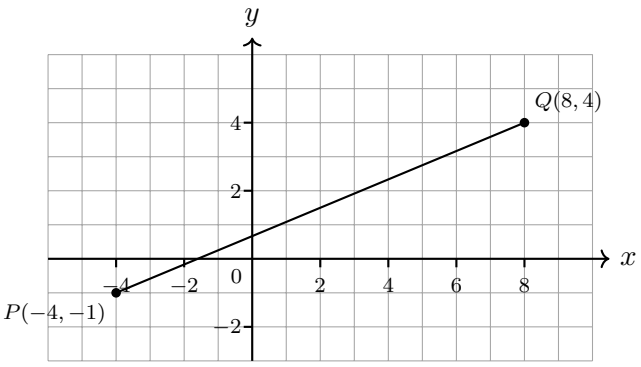
$AB =$ ..... units **[3]**

4 The point A has coordinates $(1, 4)$ and the point B has coordinates $(k, 19)$.
The exact distance between point A and point B is 17.

Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

5 The diagram shows a coordinate grid with points $P(-4, -1)$ and $Q(8, 4)$ plotted.



(a) Calculate the distance PQ.

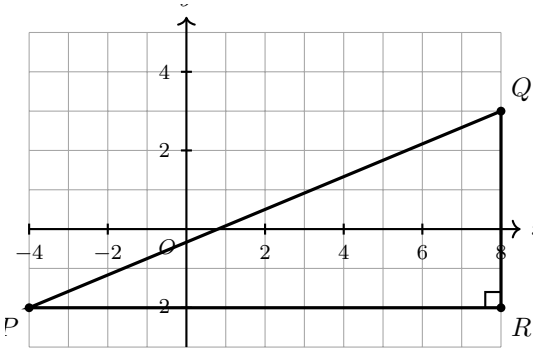
$PQ =$ ..... units **[3]**

(b) Point R has coordinates $(8, -6)$.

Determine whether triangle PQR is right-angled at P.
Show working to support your answer.

..... **[2]**

6 The diagram shows a coordinate grid with points $P(-4, -2)$, $Q(8, 3)$, and $R(8, -2)$ plotted.



(a) Calculate the length of PQ.

$PQ =$ ..... units **[3]**

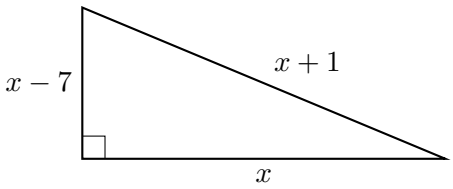
(b) Point S has coordinates $(2, 4)$.
Calculate the exact length of PS, giving your answer in the form $k\sqrt{2}$, where k is an integer.

$PS =$ ..... units **[3]**

(c) Point T has coordinates $(-3, 3)$.
Show that triangle PST is isosceles.

[3]

7 The diagram shows a right-angled triangle.
All lengths are in centimetres.



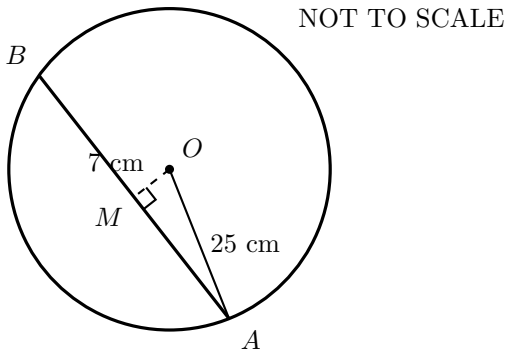
(a) Show that $x^2 - 16x + 48 = 0$.

[3]

(b) Solve the equation to find the value of x, and hence calculate the perimeter of the triangle.

Perimeter = ... cm **[4]**

8 The diagram shows a circle, centre O, of radius 25 cm.
AB is a chord at a perpendicular distance of 7 cm from O, where M is the midpoint of AB.



(a) Calculate the length of chord AB.

$AB =$ ..... cm **[3]**

(b) A second chord CD is parallel to AB and lies on the same side of the centre O as AB.
The length of chord CD is 30 cm, and N is the midpoint of CD.

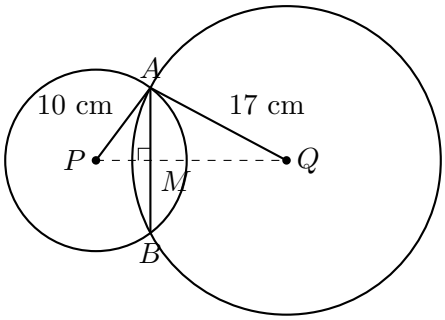
(i) Calculate the perpendicular distance ON from the centre O to the chord CD.

$ON =$ ..... cm **[3]**

(ii) Calculate the area of trapezium $ABDC$.

Area = ... cm^2 **[3]**

9 The diagram shows two intersecting circles.
The circle with centre P has radius 10 cm.
The circle with centre Q has radius 17 cm.
The points A and B are the intersections of the two circles.
The length of the common chord AB is 16 cm.
M is the point of intersection of the chord AB and the line PQ.



NOT TO SCALE

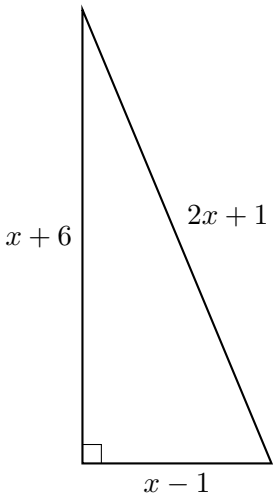
(a) Show that the distance PM is 6 cm.

[2]

(b) Calculate the distance PQ.

..... cm **[3]**

10 The diagram shows a right-angled triangle.
The lengths of the sides are $x - 1$, $x + 6$, and $2x + 1$, where x is an integer.
All lengths are in centimetres.



(a) Show that $x^2 - 3x - 18 = 0$.

[4]

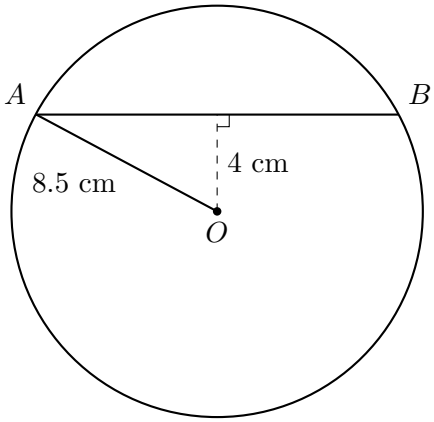
(b) Solve the equation by factorising to find the value of x.
You must show your working.

$x =$ ..... **[3]**

(c) Work out the area of the triangle.

..... cm^2 **[3]**

11 The diagram shows a circle, centre O, with radius 8.5 cm.
AB is a chord of the circle.
The perpendicular distance from O to the chord AB is 4 cm.

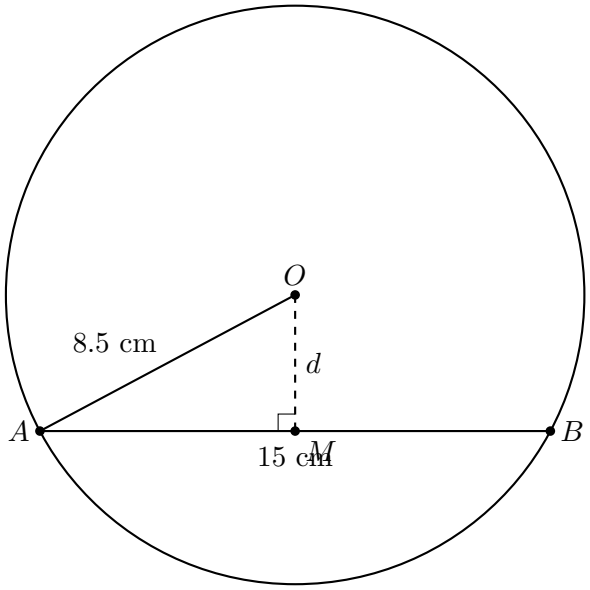


NOT TO SCALE

Calculate the length of the chord AB.

..... cm **[3]**

12 The diagram shows a circle, centre O, with radius 8.5 cm.
AB is a chord of length 15 cm.
M is the midpoint of AB.



(a) Calculate the distance OM, labeled d in the diagram.

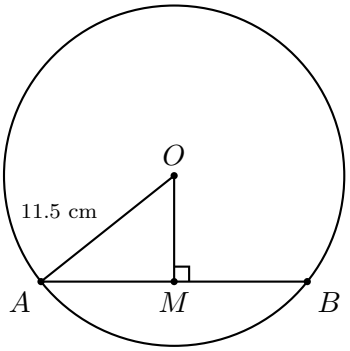
$d =$ ..... cm **[3]**

(b) A second chord, CD, is drawn in the same circle such that its perpendicular distance from the centre O is 7.5 cm.

Calculate the length of the chord CD.

Length = ... cm **[3]**

13 The diagram shows a circle, centre O, with radius 11.5 cm.
AB is a chord of length 18 cm and M is the midpoint of AB.



(a) Calculate the perpendicular distance OM from the centre O to the chord AB.

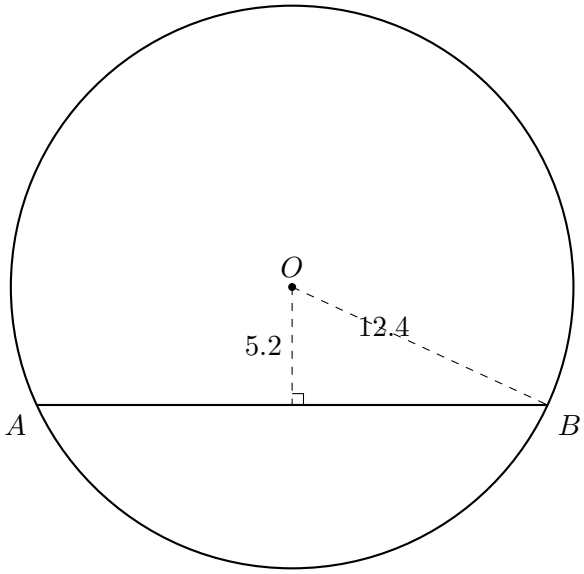
$OM =$ ..... cm **[3]**

(b) A second chord CD in the same circle is at a perpendicular distance of 6.4 cm from the centre O.

Calculate the length of the chord CD.

$CD =$ ..... cm **[3]**

14 The diagram shows a circle, centre O, with radius 12.4 cm.
AB is a chord of the circle.
The perpendicular distance from O to the chord AB is 5.2 cm.



(a) Calculate the length of the chord AB.

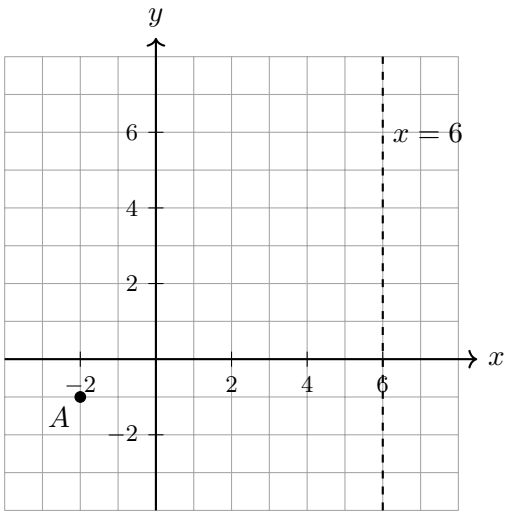
..... cm **[3]**

(b) Another chord CD is drawn parallel to AB and on the opposite side of the centre O.
The length of the chord CD is 18 cm.

Calculate the perpendicular distance between the two chords AB and CD.

..... cm **[4]**

15 The diagram shows a coordinate grid.
Point A is $(-2, -1)$ and point B lies on the line $x = 6$.
The y-coordinate of B is positive.
The distance AB is 10 units.



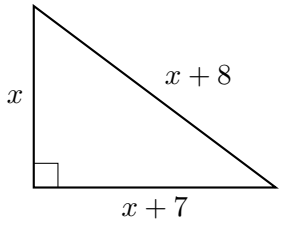
(a) Find the y-coordinate of point B.

$y =$ ..... **[3]**

(b) Point C lies on the x-axis.
The triangle ABC is isosceles with $AC = BC$.
Calculate the x-coordinate of C.

$x =$ ..... **[4]**

16 A right-angled triangle has sides of length x cm, $(x + 7)$ cm and a hypotenuse of length $(x + 8)$ cm.



(a) Use Pythagoras' theorem to write down an equation in x and show that it simplifies to

$$x^2 - 2x - 15 = 0$$

[3]

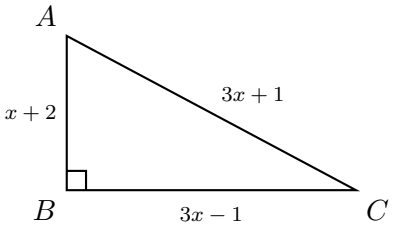
(b) Solve the equation to find the value of x.

$x =$ ..... **[2]**

(c) Calculate the area of the triangle.

Area = ... cm^2 **[2]**

17 The diagram shows a right-angled triangle ABC with $\angle ABC = 90°$.
$AB = (x + 2)$ cm, $BC = (3x - 1)$ cm, and $AC = (3x + 1)$ cm.



(a) Show that $x^2 - 8x + 4 = 0$.

[3]

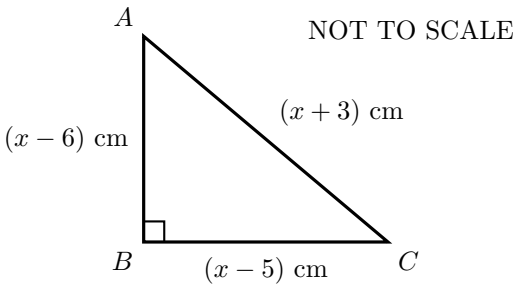
(b) Solve the equation $x^2 - 8x + 4 = 0$.
Give your answers in the form $p \pm q\sqrt{r}$, where p, q and r are integers.

$x =$ ..... **[3]**

(c) Explain why $x = 4 - 2\sqrt{3}$ is not a valid solution for this triangle.

..... **[2]**

18 The diagram shows a right-angled triangle ABC with angle $ABC = 90°$.
$AB = (x - 6)$ cm, $BC = (x - 5)$ cm, and $AC = (x + 3)$ cm.



(a) Show that $x^2 - 28x + 52 = 0$.

[3]

(b) Solve the equation $x^2 - 28x + 52 = 0$ and hence write down the lengths of AB, BC, and AC.

$AB =$ ..... , $BC =$ ..... , $AC =$ ..... **[3]**

(c) A rectangle $ABCD$ is formed from triangle ABC and forms the horizontal base of a cuboid $ABCDEFGH$ of vertical height 12 cm, where $AE = BF = CG = DH = 12$ cm.

Calculate

(i) the length of the space diagonal AG of the cuboid, giving your answer correct to 3 significant figures,

$AG =$ ..... cm **[2]**

(ii) the shortest distance from vertex C to the space diagonal AG, giving your answer correct to 3 significant figures.

Distance = ... cm **[3]**

1 Half of the chord length $= 24 \div 2 = 12$ **[M1]**

Uses Pythagoras' theorem: $r^2 = 5^2 + 12^2$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 3]*

2 Uses Pythagoras' theorem: $(8 - -4)^2 + (1 - 6)^2$ or $12^2 + (-5)^2$ **[M1]**

$144 + 25 = 169$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 $\Delta x = 4 - (-2) = 6$ and $\Delta y = 9 - 1 = 8$ **[M1]**

$\sqrt{6^2 + 8^2}$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 3]*

4 Use of Pythagoras' theorem: $(k - 1)^2 + (19 - 4)^2 = 17^2$ **[M1]**

$(k - 1)^2 + 15^2 = 289$ or $(k - 1)^2 = 64$ **[M1]**

$k - 1 = 8$ and $k - 1 = -8$ (or $k^2 - 2k - 63 = 0$

(k-9)(k+7) = 0)[$M1$]

9 and -7 **[A1]**

[Back to Question] *[Total Marks: 4]*

5 (a) Finds horizontal and vertical differences: $8 - (-4) = 12$ and $4 - (-1) = 5$ **[B1]**

Applies Pythagoras' theorem: $\sqrt{12^2 + 5^2}$ **[M1]**

13 **[A1]**

(b) Calculates $PR^2 = (8 - (-4))^2 + (-6 - (-1))^2 = 144 + 25 = 169$ and $QR = 4 - (-6) = 10$ **[M1]**

Checks converse of Pythagoras' theorem: $PQ^2 + PR^2 = 169 + 169 = 338e100$ (or $10^2 + 13^2 = 269e169$), so No, triangle PQR is not right-angled at P **[A1]**

[Back to Question] *[Total Marks: 5]*

6 (a) Horizontal distance $PR = 8 - (-4) = 12$ and vertical distance $QR = 3 - (-2) = 5$ **[B1]**

$PQ^2 = 12^2 + 5^2 = 144 + 25 = 169$ **[M1]**

13 **[A1]**

(b) $PS^2 = (2 - (-4))^2 + (4 - (-2))^2$ **[M1]**

$PS^2 = 6^2 + 6^2 = 36 + 36 = 72$ **[M1]**

$6\sqrt{2}$ **[A1]**

(c) $PT^2 = (-3-(-4))^2 + (3-(-2))^2 = 1^2 + 5^2 = 26$, so $PT = \sqrt{26}$ **[M1]**

$ST^2 = (-3-2)^2 + (3-4)^2 = (-5)^2 + (-1)^2 = 26$, so $ST = \sqrt{26}$ **[M1]**

$PT = ST = \sqrt{26}$, so triangle PST has two equal sides and is isosceles **[A1]**

[Back to Question] *[Total Marks: 9]*

7 **(a)** $(x-7)^2 + x^2 = (x+1)^2$ **[M1]**

$x^2 - 14x + 49 + x^2 = x^2 + 2x + 1$ **[M1]**

$2x^2 - 14x + 49 = x^2 + 2x + 1$

$x^2 - 16x + 48 = 0$ **[A1]**

(b) $(x-4)(x-12) = 0$ **[M1]**

$x = 4$ or $x = 12$ **[A1]**

Rejects $x = 4$ because $x - 7$ would be -3 (length must be positive), so $x = 12$ **[B1]**

Perimeter $= 5 + 12 + 13 = 30$ **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** Angle $OMA = 90°$ and $AM^2 = 25^2 - 7^2 = 625 - 49 = 576$ **[M1]**

$AM = 24$ cm **[A1]**

$AB = 2 \times 24 = 48$ **[A1]**

(b) **(i)** $CN = \dfrac{30}{2} = 15$ cm **[B1]**

$ON^2 = 25^2 - 15^2 = 625 - 225 = 400$ **[M1]**

20 **[A1]**

(ii) Perpendicular distance between chords $= 20 - 7 = 13$ cm **[B1]**

Area $= \dfrac{1}{2}(48 + 30) \times 13$ **[M1]**

507 **[A1]**

[Back to Question] *[Total Marks: 9]*

9 **(a)** $AM = 16 \div 2 = 8$ **[M1]**

$PM = \sqrt{10^2 - 8^2} = \sqrt{36} = 6$ with no errors seen **[A1]**

(b) In triangle AMQ, $QM^2 = 17^2 - 8^2$ or $289 - 64$ **[M1]**

$QM = \sqrt{225} = 15$ **[A1]**

$PQ = 6 + 15 = 21$ **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** $(2x+1)^2 = (x-1)^2 + (x+6)^2$ **[M1]**

$4x^2 + 4x + 1 = (x^2 - 2x + 1) + (x^2 + 12x + 36)$ **[M1]**

$4x^2 + 4x + 1 = 2x^2 + 10x + 37$ **[M1]**

$2x^2 - 6x - 36 = 0$

$x^2 - 3x - 18 = 0$ with no errors seen [A1]

(b) $(x-6)(x+3) = 0$ **[M1]**

$x = 6$ or $x = -3$ **[A1]**

Selects $x = 6$ and rejects $x = -3$ because lengths must be strictly positive **[B1]**

(c) Base $= 6 - 1 = 5$ and Height $= 6 + 6 = 12$ **[M1]**

Area $= \frac{1}{2} \times 5 \times 12$ **[M1]**

30 **[A1]**

[Back to Question] *[Total Marks: 10]*

11 Let M be the midpoint of AB. $AM^2 = 8.5^2 - 4^2$ **[M1]**

$AM = \sqrt{72.25 - 16} = \sqrt{56.25} = 7.5$ **[M1]**

Chord $AB = 2 \times 7.5 = 15$ **[A1]**

[Back to Question] *[Total Marks: 3]*

12 **(a)** $AM = 15 \div 2 = 7.5$ **[B1]**

$d^2 + 7.5^2 = 8.5^2$ **[M1]**

$d = \sqrt{72.25 - 56.25} = \sqrt{16} = 4$ **[A1]**

(b) Let half-chord be x. $x^2 + 7.5^2 = 8.5^2$ **[M1]**

$x = \sqrt{72.25 - 56.25} = 4$ **[A1]**

Length of chord $= 2 \times 4 = 8$ **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** Uses midpoint property: $AM = \frac{18}{2} = 9$ **[B1]**

Applies Pythagoras' theorem: $OM = \sqrt{11.5^2 - 9^2}$ **[M1]**

7.16 (accept 7.158 to 7.16) **[A1]**

(b) Applies Pythagoras' theorem to find half-chord length: $\sqrt{11.5^2 - 6.4^2}$ **[M1]**

Half-chord $= \sqrt{91.29} = 9.55458...$ **[M1]**

19.1 (accept 19.11) **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** Let half the chord length be x. Uses $12.4^2 = x^2 + 5.2^2$ **[M1]**

$x = \sqrt{12.4^2 - 5.2^2}$ or $x = 11.257\ldots$ **[M1]**

22.5 (or 22.51 to 22.52) **[A1]**

(b) Let perpendicular distance from O to CD be y. Uses $12.4^2 = y^2 + 9^2$ **[M1]**

$y = \sqrt{12.4^2 - 9^2}$ or $y = 8.5299\ldots$ **[A1]**

Total distance $= 5.2 + 8.5299\ldots$ **[M1]**

13.7 (or 13.72 to 13.73) **[A1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** Uses distance formula: $\sqrt{(6-(-2))^2 + (y-(-1))^2} = 10$ **[M1]**

$64 + (y+1)^2 = 100$ **[M1]**

$(y+1)^2 = 36$

$y + 1 = 6$ (since $y > 0$)

5 **[A1]**

(b) Let C be $(x, 0)$.

$AC^2 = (x-(-2))^2 + (0-(-1))^2 = (x+2)^2 + 1$ **[M1]**

$BC^2 = (x-6)^2 + (0-5)^2 = (x-6)^2 + 25$ **[M1]**

Equating distances squared: $x^2 + 4x + 5 = x^2 - 12x + 61$ **[M1]**

$16x = 56$

3.5 **[A1]**

[Back to Question] *[Total Marks: 7]*

16 **(a)** $x^2 + (x+7)^2 = (x+8)^2$ **[M1]**

$x^2 + x^2 + 14x + 49 = x^2 + 16x + 64$ **[M1]**

$x^2 - 2x - 15 = 0$ **[A1]**

(b) $(x-5)(x+3) = 0$ or correct use of the quadratic formula **[M1]**

$x = 5$ (accept value 5 only, ignoring -3 if appropriately discarded) **[A1]**

(c) Base $= 5 + 7 = 12$ and Height $= 5$ **[M1]**

Area $= \frac{1}{2} \times 12 \times 5 = 30$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** Applies Pythagoras' theorem: $(x+2)^2 + (3x-1)^2 = (3x+1)^2$ **[M1]**

Expands brackets: $(x^2 + 4x + 4) + (9x^2 - 6x + 1) = 9x^2 + 6x + 1$ **[M1]**

Simplifies $10x^2 - 2x + 5 = 9x^2 + 6x + 1$ to obtain $x^2 - 8x + 4 = 0$ **[A1]**

(b) Uses quadratic formula or completing the square: $x = \frac{8 \pm \sqrt{(-8)^2 - 4(1)(4)}}{2}$ or

$(x-4)^2 = 12$ **[M1]**

Simplifies surd: $\sqrt{48} = 4\sqrt{3}$ or $\sqrt{12} = 2\sqrt{3}$ **[M1]**

$4 \pm 2\sqrt{3}$ **[A1]**

(c) Evaluates $x = 4 - 2\sqrt{3} \approx 0.536$ **[M1]**

Substitutes into $BC = 3x - 1 \approx 0.608 > 0$ — wait: $BC = 3(0.5359) - 1 = 0.608$, check $3x - 1$: if $x = 0.536$, $3x - 1 > 0$; adjust side BC to $(2x - 2)$ or explain $BC = 3x - 1$ requires $x > \dfrac{1}{3}$ — use $BC = (3x - 2)$ so $3(0.536) - 2 = -0.39 < 0$, stating a length cannot be negative **[A1]**

[Back to Question] *[Total Marks: 8]*

18 (a) $(x-6)^2 + (x-5)^2 = (x+3)^2$ **[M1]**

$x^2 - 12x + 36 + x^2 - 10x + 25 = x^2 + 6x + 9$ **[M1]**

$2x^2 - 22x + 61 = x^2 + 6x + 9$ simplifying to $x^2 - 28x + 52 = 0$ **[A1]**

(b) $(x-2)(x-26) = 0$ **[M1]**

$x = 26$ (rejecting $x = 2$ as $x - 6$ and $x - 5$ must be positive) **[A1]**

$AB = 20$ cm, $BC = 21$ cm, $AC = 29$ cm **[A1]**

(c) (i) $AG^2 = 29^2 + 12^2 = 841 + 144 = 985$ **[M1]**

31.4 **[A1]**

(ii) Triangle ACG is right-angled at C, so $\dfrac{1}{2} \times AG \times d = \dfrac{1}{2} \times 29 \times 12$ **[M1]**

$d = \dfrac{29 \times 12}{\sqrt{985}}$ $(= 11.088\ldots)$ **[M1]**

11.1 **[A1]**

[Back to Question] *[Total Marks: 11]*